Cells, biological molecules and movement of substances

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 | Level of organisation | [1]

Answer: B

Explanation

Organelles (such as mitochondria) are inside cells; similar cells form tissues; different tissues form organs; organs work together in systems. A puts the cell before the organelle, but organelles are parts of cells. C puts tissue before cell, but tissues are made of cells. D puts organ before tissue, but organs are made of tissues.

2 · Part 1

Paper 1 & 2 | Cell structure | [1]

Answer: D

Explanation

The cell membrane controls which substances enter and leave the cell. A is wrong because cell sap is found in the vacuole; the cell wall supports the cell and stops it bursting. B is wrong because genetic information is stored in the nucleus; ribosomes are where proteins are made. C is wrong because chloroplasts absorb light energy for photosynthesis; the vacuole contains cell sap and helps keep the cell firm.

3 · Part 1

Paper 2 only | Cell structure | [1]

Answer: A

Explanation

One risk of using embryonic stem cells is that they may keep dividing out of control and form a tumour, so A is a disadvantage. B, C and D are all advantages: being able to become many types of cell, replacing damaged cells and being easy to grow in large numbers are the reasons doctors want to use stem cells.

Topical Hub
topicalhub-pages.dev

4 · Part 1 Paper 1 & 2 Biological molecules [1]

Answer: C

Explanation

Carbohydrates and lipids contain only carbon, hydrogen and oxygen. Proteins contain carbon, hydrogen, oxygen and nitrogen. So nitrogen is in all proteins but not in lipids. A, B and D are found in both proteins and lipids.

5 · Part 1 Paper 1 & 2 Biological molecules [1]

Answer: B

Explanation

Benedict's solution turning brick red on heating shows glucose. Biuret solution turning purple shows protein. Iodine solution staying orange-brown shows that starch is absent (starch would turn it blue-black). So the food contains glucose and protein only. A and D include starch, which was not detected. C leaves out glucose and includes starch.

6 · Part 1 Paper 1 & 2 Biological molecules [1]

Answer: D

Explanation

At low temperature the enzyme and substrate molecules have less kinetic energy, so they move more slowly and collide less often. Fewer enzyme–substrate complexes form, so the rate is slow. A is wrong because enzymes are molecules (proteins), not living things, so they cannot be killed; low temperature slows them down without damaging them. B is wrong because enzymes are catalysts and are not used up. C is wrong because low temperature does not change the shape of the substrate.

7 · Part 1 Paper 2 only Biological molecules [1]

Answer: C

Explanation

The buffer solution keeps the pH of the mixture at a set value, so each test is carried out at the pH being investigated. A is wrong because the substrate is added separately. B is wrong because the buffer does not speed up the reaction; its job is to fix the pH. D is wrong because a buffer does not provide energy for the reaction; its only job is to fix the pH.

Topical Hub
topical-hub.pages.dev

8 · Part 1

Paper 1 & 2 | Movement of substances into and out of cells | [1]

Answer: A

Explanation

Glucose is moving from a lower concentration (in the small intestine) to a higher concentration (inside the cells), which is against its concentration gradient. This needs active transport, which uses energy from respiration. B is wrong because glucose is not moving down its gradient. C is wrong because osmosis is the movement of water, not glucose. D is wrong because diffusion is passive and does not use energy from respiration.

9 · Part 1

Paper 1 & 2 | Level of organisation | [5]

Mark scheme

(a) **[2]**

- all five correct: chloroplast organelle, red blood cell cell, xylem tissue, leaf organ, digestive system system (2)
- three or four correct (1)

(b) **[1]**

- a group of similar cells that work together to carry out a particular function (1)

ALLOW group of cells of the same type with the same function

(c) **[2]**

- it is made of several different tissues (1)
- (the tissues) work together to carry out a particular function, e.g. digestion of food (1)

Topical Hub
topical-hub.pages.dev

Worked solution

(a) A chloroplast is a structure inside a cell, so it is an organelle. A red blood cell is a single cell. Xylem is made of many similar cells working together to carry water, so it is a tissue. A leaf contains several tissues (for example xylem and the cells that photosynthesise), so it is an organ. The digestive system is a group of organs, so it is a system.

(b) A tissue is a group of similar cells that work together to carry out a particular function.

(c) The stomach contains different types of tissue (muscle, glandular and epithelial). These tissues work together to do one job, digesting food, so the stomach is an organ.

10 · Part 1 Paper 1 & 2 Cell structure [6]

Mark scheme

(a) **[2]**

- leaf cell: yes, yes, yes, yes (1)
- cheek cell: no, no, yes, no (1)

(b) **[2]**

- contains the genetic material OR contains DNA OR contains chromosomes OR contains genes (1)
- controls the activities of the cell (1)

ALLOW controls what proteins are made

(c) **[2]**

- roots are underground OR roots receive no light (1)
- so root cells cannot photosynthesise, so chloroplasts are not needed (1)

Worked solution

(a) Plant leaf cells have a cell wall, chloroplasts, a nucleus and a large vacuole containing cell sap. Animal cells such as cheek cells have a nucleus, but no cell wall, no chloroplasts and no large vacuole. Both cells also have cytoplasm, a cell membrane and mitochondria.

(b) The nucleus contains the genetic material (DNA in chromosomes). This controls the activities of the cell.

(c) Chloroplasts absorb light for photosynthesis. Roots grow in the soil where there is no light, so photosynthesis cannot happen there and chloroplasts would be of no use.

Topical Hub
topicalhubpages.dev

11 · Part 1 Paper 1 & 2 Cell structure [6]

Mark scheme

(a) **[1]**

- 20 (times) (1)

(b) **[3]**

- heart muscle contracts all the time OR contracts continuously (1)
- (so heart muscle cells) need a lot of energy (1)
- mitochondria are where aerobic respiration takes place, which releases energy (1)

ALLOW respiration produces ATP for the second or third marking point

REJECT respiration makes energy OR produces energy

(c) **[2]**

- ribosomes (1)
- proteins are made at ribosomes OR ribosomes are the site of protein synthesis (1)

ALLOW rough endoplasmic reticulum for the first marking point

Worked solution

(a) 5000 ÷ 250 = 20, so a heart muscle cell has 20 times as many mitochondria as a skin cell. Check: 250 × 20 = 5000.

(b) The heart muscle contracts all the time to pump blood, so its cells need a lot of energy. Aerobic respiration takes place in mitochondria and releases energy (as ATP). Having many mitochondria means a lot of energy can be released.

(c) The enzymes are proteins, and proteins are made at ribosomes. Cells that make large amounts of protein need very many ribosomes.

Topical Hub
topical-hub-pages.dev

12 · Part 1 Paper 2 only Cell structure [6]

Mark scheme

(a) **[1]**

- the process by which a cell becomes specialised for a particular function (1)

ALLOW an unspecialised cell develops into a specific type of cell

(b) *(any three)* **[3]**

- differentiation produces specialised cells (1)
- specialised cells have structures or features adapted to a particular function (1)
- different cells can carry out different functions OR the organism can carry out many functions (1)
- so each function is carried out efficiently (1)
- specialised cells can form tissues and organs (1)

(c) **[2]**

- the cells can divide to make new cells that can differentiate into different types of specialised cell (1)
- so the plant can keep growing (roots and shoots) throughout its life OR can replace damaged parts (1)

Worked solution

(a) Cell differentiation is the process by which a cell becomes specialised for a particular function.

(b) A multicellular organism must do many jobs, such as carrying oxygen, contracting and sending impulses. Differentiation produces specialised cells, each with features suited to one job. Because cells are specialised, each job is done efficiently, and groups of specialised cells form the tissues and organs of the body.

(c) The unspecialised cells keep dividing, and the new cells can differentiate into whatever type of cell is needed. This allows the roots and shoots to keep growing for the whole life of the plant and to replace damaged parts.

Topical Hub
topical-hub.pages.dev

13 · Part 1 Paper 2 only Cell structure [7]

Mark scheme

(a) **[2]**

- embryonic stem cells can become almost any type of cell, but adult stem cells from bone marrow can become only a few types, mainly blood cells (1)
- so embryonic stem cells could replace damaged cells in many different tissues or organs (1)

ALLOW a named example of a non-blood cell, e.g. nerve cells or muscle cells, for the second marking point

(b) **[5]**

Indicative content (any five, max 4 from advantages only or disadvantages only):

Advantages of embryonic stem cells:

- they can become almost any type of cell, so more diseases can be treated (1)
- the patient does not need a needle pushed into a bone OR avoids a painful procedure for the patient (1)
- the embryos are already available as they are left over from fertility treatment (1)

Disadvantages of embryonic stem cells:

- the embryo is destroyed (1)
- some people think this is wrong OR ethical objections because the embryo could have developed into a person (1)
- the cells are not the patient's own, so they may be rejected by the patient's immune system (1)
- the patient may need drugs to stop rejection (1)
- the cells may divide in an uncontrolled way and form a tumour OR cancer (1)

ALLOW comparison points about bone marrow cells, e.g. bone marrow cells are not rejected because they are the patient's own cells, credited as a disadvantage of embryonic cells

Worked solution

(a) The table shows that embryonic stem cells can become almost any type of cell, but bone marrow stem cells can become only a few types, mainly blood cells. So embryonic stem cells could be used to replace damaged cells in many tissues, for example nerve or muscle, and so could treat more diseases.

(b) A good answer weighs both sides. For embryonic stem cells: they can become almost any type of cell, the embryos are left over from fertility treatment, and the patient avoids having a needle pushed into a bone. Against embryonic stem cells: the embryo is destroyed, which many people think is wrong because the embryo could have become a person; the cells come from another individual, so the patient's immune system may reject them and the patient may need drugs to prevent rejection; and stem cells may divide out of control and form a tumour. Bone marrow cells come from the patient, so they are not rejected and raise fewer ethical concerns, but they can only become a few types of cell.

Topical Hub
topical-hub.pages.dev

14 · Part 1 Paper 2 only Cell structure [8]

Mark scheme

(a) **[2]**

- (45 − 38) ÷ 38 × 100 OR 7 ÷ 38 × 100 (1)
- 18 % (1)

ALLOW 18.4 % OR 18.42 %

(b) **[2]**

- as a control OR for comparison (1)
- to show that any improvement in group 1 was caused by the stem cells and not by the injection itself OR not by natural recovery over time (1)

(c) **[2]**

- the stem cells differentiated into (heart) muscle cells (1)
- these replaced the dead or damaged muscle cells, so more muscle could contract (1)

ALLOW the stem cells divided by mitosis to make more cells for the first marking point

(d) **[2]**

- support: group 1 improved more than group 2 (7 compared with 2) (1)
- against, any one: only ten patients in each group OR the improvement was small OR group 2 also improved OR the hearts were still not working at a normal level OR only measured for six months, so it is not a cure (1)

Worked solution

(a) Increase = 45 − 38 = 7. Percentage increase = 7 ÷ 38 × 100 = 18.4, which is 18 % to 2 significant figures. Check: 38 + 18 % of 38 = 38 + 6.8 = 44.8, which is about 45.

(b) Group 2 is a control. Patients may improve just because time passes, or because of the injection itself. Comparing with group 2 shows whether the extra improvement in group 1 was due to the stem cells.

(c) The stem cells could differentiate into heart muscle cells. These would replace the muscle cells killed by the heart attack, so more of the heart muscle could contract and more blood would be pumped out with each beat.

(d) The data give some support: group 1 improved by 7 but group 2 by only 2. But the claim is too strong: there were only ten patients in each group, the improvement was small, group 2 also improved, and the patients were only followed for six months. The trial shows a possible benefit, not a cure.

Topical Hub
topical-hub.pages.dev

15 · Part 1 Paper 1 & 2 Biological molecules [5]

Mark scheme

(a) **[2]**

- starch and glycogen: simple sugars OR glucose (1)
- protein: amino acids (1)

(b) **[1]**

- carbon, hydrogen and oxygen (all three needed) (1)

(c) *(any two)* **[2]**

- glycogen does not change the concentration of the cell contents OR does not affect the movement of water into or out of the cell by osmosis (1)
- glycogen cannot diffuse out of the cell OR glycogen is too large to leave the cell (1)
- many glucose units can be stored in a small space (1)

Worked solution

(a) Starch and glycogen are both made from many simple sugar (glucose) units joined together. Proteins are made from amino acids.

(b) Starch is a carbohydrate, and carbohydrates contain carbon, hydrogen and oxygen.

(c) Glucose dissolves, so storing a lot of it would make the cell contents very concentrated and water would move into the cell by osmosis. Glycogen is insoluble, so it does not affect osmosis. Glycogen is also a large molecule, so it stays inside the cell and cannot diffuse out.

Topical Hub
topical-hub-pages.dev

16 · Part 1 Paper 1 & 2 Biological molecules [6]

Mark scheme

(a) **[3]**

- add ethanol to the crushed peanuts and shake (1)
- pour the ethanol into water OR add water to the ethanol (1)
- a cloudy white emulsion forms if fat is present (1)

ALLOW milky white for cloudy white

(b) **[2]**

- as a control OR for comparison (1)
- to show that the result is caused by fat in the peanuts and not by the ethanol or water alone (1)

(c) **[1]**

- keep the ethanol away from flames OR do not use a Bunsen burner near the ethanol (1)

ALLOW wear eye protection

Worked solution

(a) Fat dissolves in ethanol. Shake the crushed peanuts with ethanol so any fat dissolves. Then pour the ethanol into water. Fat does not dissolve in water, so it forms tiny droplets that make a cloudy white emulsion. If there is no fat, the mixture stays clear.

(b) The test with water instead of peanuts is a control. If the water tube stays clear and the peanut tube goes cloudy white, the cloudiness must be due to fat from the peanuts.

(c) Ethanol catches fire easily, so there must be no naked flames nearby.

Topical Hub
topicalhub.ags.dev

17 · Part 1 Paper 1 & 2 Biological molecules [5]

Mark scheme

(a) **[1]**

- a substance that speeds up a chemical reaction without being used up OR without being changed (1)

(b) **[2]**

- fatty acids (1)
- glycerol (1)

(c) **[2]**

- the active site of the protease has a specific shape that matches its substrate, protein (1)
- lipid molecules do not fit the active site, so no enzyme–substrate complex forms (1)

ALLOW active site is complementary to protein OR not complementary to lipid

Worked solution

(a) A catalyst speeds up a chemical reaction but is not used up or changed by the reaction. Enzymes are proteins that act as catalysts in living things, so they are called biological catalysts.

(b) Lipids are made of fatty acids and glycerol, so these are the products when lipase digests lipids.

(c) Each enzyme has an active site with a particular shape. Only a substrate with a matching shape can fit into it. The protease's active site matches protein molecules, so lipid molecules do not fit and are not digested.

Topical Hub
topical-hub.pages.dev

18 · Part 1 Paper 1 & 2 Biological molecules [6]

Mark scheme

(a) **[2]**

- the rate increases from 20 °C to 70 °C (1)
- then decreases between 70 °C and 80 °C (1)

ALLOW data quoted, e.g. rises from 4 to 52 then falls to 30, for either marking point

(b) **[1]**

- human enzyme 40 °C and hot-spring enzyme 70 °C (both needed) (1)

(c) **[3]**

- the high temperature changes the shape of the active site (1)
- the substrate no longer fits the active site (1)
- no enzyme–substrate complexes form OR the enzyme is denatured (1)

REJECT the enzyme is killed

Worked solution

(a) For the hot-spring enzyme the rate rises steadily from 4 at 20 °C to a peak of 52 at 70 °C. Above this it falls, to 30 at 80 °C.

(b) The highest rate for the human enzyme is 38 at 40 °C. The highest rate for the hot-spring enzyme is 52 at 70 °C.

(c) At 70 °C the human enzyme has been denatured. The high temperature changes the shape of its active site, so the substrate no longer fits. No enzyme–substrate complexes can form, so there is no reaction.

Topical Hub
topical-hub.pages.dev

19 · Part 1

Paper 1 & 2 Biological molecules [7]

Mark scheme

(a) **[2]**

- so that the solutions reach the temperature of the water bath (1)
- so the reaction takes place at the temperature being tested OR to control the temperature (1)

(b) **[1]**

- 81 s (1)

(c) **[2]**

- 95 s (1)
- 63 s (1)

(d) **[2]**

- test more temperatures between 40 °C and 60 °C (1)
- at smaller intervals, e.g. every 2 °C OR every 5 °C (1)

ALLOW repeat each temperature more times only with the first marking point

Worked solution

(a) Each solution needs time to warm or cool to the water bath temperature. If they were mixed straight away, the reaction would start at room temperature, and the temperature being tested would not be the real temperature of the reaction.

(b) Mean = (84 + 78 + 81) ÷ 3 = 243 ÷ 3 = 81 s. Check: 81 is between 78 and 84, as a mean must be.

(c) The 95 s result is much longer than the other two results at 50 °C (62 s and 64 s), so it is anomalous. Mean without it = (62 + 64) ÷ 2 = 63 s.

(d) The shortest mean time was at 50 °C, but the real optimum could be anywhere between 40 °C and 60 °C. Hana should test more temperatures in this range, at small intervals such as every 2 °C, and repeat each one.

Topical Hub
topical-hub.pages.dev

20 · Part 1 Paper 1 & 2 Biological molecules [6]

Mark scheme

(a) **[2]**

- activity is highest at pH 2 (1)
- activity decreases as the pH increases above 2, to zero at pH 8 and pH 10 (1)

ALLOW activity is lower at pH 1 than at pH 2

(b) **[3]**

- the pH changes the shape of the active site (1)
- the substrate OR protein no longer fits the active site (1)
- no enzyme–substrate complexes form OR enzyme X is denatured (1)

(c) **[1]**

- enzyme Y has no activity at pH 2 OR enzyme Y works best at pH 8 (1)

Worked solution

(a) Enzyme X is most active at pH 2 (48 units). Its activity is lower at pH 1, and above pH 2 it falls (20 at pH 4, 2 at pH 6) until it is 0 at pH 8 and pH 10.

(b) At pH 8 the shape of the active site of enzyme X has changed. The protein substrate no longer fits, so no enzyme–substrate complexes form and there is no activity.

(c) Enzyme Y has no activity at pH 2 and works best at pH 8, so the small intestine must have a higher pH than the stomach for enzyme Y to digest protein.

Topical Hub
topical-hub.pages.dev

21 · Part 1 Paper 2 only Biological molecules [7]

Mark scheme

(a) **[1]**

- volume of oxygen (collected in 2 minutes) (1)

(b) **[2]**

- 30 ÷ 2 (1)
- 15 cm^3 per minute (1)

(c) **[2]**

- most oxygen was collected at pH 7, so pH 7 was the best of the pH values tested (1)
- but only whole-number pH values were tested, so the optimum could be anywhere between pH 6 and pH 8 OR only one test at each pH, so the result may not be repeatable (1)

(d) **[2]**

- repeat the test with water instead of catalase OR with boiled catalase, keeping everything else the same (1)
- little or no oxygen is collected (1)

Worked solution

(a) Leo changed the pH (independent variable) and measured the volume of oxygen collected in 2 minutes (dependent variable).

(b) Rate = 30 ÷ 2 = 15 cm^3 per minute. Check: 15 cm^3 per minute for 2 minutes gives 30 cm^3.

(c) pH 7 gave the most oxygen, so it was the best pH Leo tested. But he only tested whole-number pH values, and pH 8 gave nearly as much oxygen as pH 7, so the true optimum could be between pH 6 and pH 8, for example pH 7.3. He also did only one test at each pH, so he cannot be sure the results are repeatable. The conclusion that the optimum is exactly pH 7 is not justified.

(d) Leo should set up a control with water or boiled (denatured) catalase in place of the catalase, with everything else the same. If little or no oxygen is collected, the oxygen in the real tests must have come from the action of catalase.

Topical Hub
topical-hub.pages.dev

22 · Part 1 Paper 2 only Biological molecules [7]

Mark scheme

(a) **[6]**

Indicative content (any six):

- add the same volume of buffer solution to the same volume of starch solution in a test tube (1)
- use a different pH buffer for each test: pH 3, 5, 7, 9 and 11 (1)
- leave the tubes of starch and buffer and of amylase in the water bath for a few minutes before mixing so they reach 35 °C (1)
- add the same volume and concentration of amylase each time and start the stop clock (1)
- place drops of iodine solution in the spotting tile (1)
- at regular intervals, e.g. every 30 s, add a drop of the mixture to a drop of iodine solution (1)
- record the time when the iodine solution stays orange-brown OR no longer turns blue-black (1)
- repeat at each pH and calculate a mean time (1)
- the shorter the time, the higher the activity of amylase OR calculate rate as 1 ÷ time (1)

ALLOW other valid control variables, e.g. same concentration of starch solution

ALLOW any workable way of transferring a sample to the iodine solution, e.g. a dropping pipette, a syringe or a glass rod

(b) **[1]**

- wear eye protection OR avoid getting iodine solution on the skin (1)

Worked solution

(a) Put the same volume of starch solution and the same volume of a buffer (for example pH 3) into a test tube. Put amylase solution in another tube. Stand both tubes in the 35 °C water bath for a few minutes so they reach the same temperature. Put single drops of iodine solution in the wells of the spotting tile. Add the amylase to the starch and buffer and start the stop clock. Every 30 seconds, put a drop of the mixture onto a fresh drop of iodine solution. While starch is present the iodine turns blue-black; record the time when it stays orange-brown. Repeat with each of the other buffers, keeping the volume and concentration of amylase and starch and the temperature the same. Do each pH three times and calculate a mean. The pH with the shortest time is the pH at which amylase is most active.

(b) Wear eye protection when handling the solutions.

Topical Hub
topical-hub.pages.dev

23 · Part 1

Paper 1 & 2 | Movement of substances into and out of cells | [6]

Mark scheme

(a) **[2]**

- net movement of particles OR molecules (1)
- from a region of higher concentration to a region of lower concentration OR down a concentration gradient (1)

(b) **[2]**

- by osmosis (1)
- water moves from the dilute pond water into the more concentrated cytoplasm through the partially permeable cell membrane (1)

ALLOW from high water concentration to low water concentration OR water potential arguments

(c) **[2]**

- water would continue to enter the cell (by osmosis) and the cell would swell (1)
- the cell would burst because it has no cell wall (1)

Worked solution

(a) Diffusion is the net movement of particles from a region where they are at a higher concentration to a region where they are at a lower concentration, down a concentration gradient. It does not need energy from respiration.

(b) The cytoplasm is more concentrated than the pond water. Water moves by osmosis from the dilute solution (pond water) to the more concentrated solution (cytoplasm) across the partially permeable cell membrane.

(c) If the contractile vacuole stopped, water would keep entering by osmosis but would not be removed. The cell would swell. With no cell wall to resist the pressure, the cell would burst.

Topical Hub
topical-hub.pages.dev

24 · Part 1

Paper 1 & 2 Movement of substances into and out of cells [7]

Mark scheme

(a) **[2]**

- surface area = 24 (cm^2) and volume = 8 (cm^3) (1)
- 3 : 1 (1)

ALLOW 24 : 8 for the second marking point

(b) **[2]**

- the smaller the surface area : volume ratio, the longer the time taken (1)
- data quoted to support this, e.g. the 6 : 1 cube took 4 min but the 2 : 1 cube took 36 min (1)

(c) **[2]**

- the acid has a greater distance to diffuse to reach the centre of the cube (1)
- the cube has the smallest surface area : volume ratio, so there is less surface for acid to enter compared with the volume to be reached (1)

(d) **[1]**

- any one: concentration of acid, volume of acid, temperature, concentration of indicator in the agar (1)

Worked solution

(a) Surface area = $6 \times 2^2 = 6 \times 4 = 24$ cm^2. Volume = $2^3 = 8$ cm^3. Ratio = 24 : 8 = 3 : 1. Check: for a cube the ratio is 6 ÷ length of side, and 6 ÷ 2 = 3, which fits the 1 cm cube (6 : 1) and the 3 cm cube (2 : 1).

(b) As the surface area : volume ratio gets smaller, the time to become colourless gets longer. The 1 cm cube (6 : 1) took 4 minutes, the 2 cm cube (3 : 1) took 16 minutes and the 3 cm cube (2 : 1) took 36 minutes.

(c) The acid moves into the agar by diffusion. In the 3 cm cube the centre is 1.5 cm from the surface, so the acid has the furthest to travel. This cube also has the smallest surface area : volume ratio, so there is the least surface for the acid to enter for each cm^3 of agar.

(d) Keep the concentration and volume of the acid and the temperature the same, so only the size of the cube affects the time.

Topical Hub
topicalhubpages.com

25 · Part 1

Paper 1 & 2 | Movement of substances into and out of cells | [8]

Mark scheme

(a) **[2]**

- the cylinders had different initial masses (1)
- percentage change allows a fair comparison between cylinders (1)

(b) **[2]**

- (2.40 − 2.45) ÷ 2.45 × 100 (1)
- −2.0 % (1)

ALLOW −2.04 % OR 2.0 % decrease

ALLOW 2.0 % without a sign only if the working shows a decrease

(c) **[3]**

- the sucrose solution was more concentrated than the potato cell contents (1)
- water moved out of the potato cells by osmosis (1)
- through the partially permeable cell membrane, so the mass decreased (1)

ALLOW water potential arguments

(d) **[1]**

- any value from 8.0 % to 9.5 % (1)

Worked solution

(a) The cylinders did not all start with the same mass. A cylinder with a bigger starting mass could gain or lose more grams just because it is bigger. Percentage change takes the starting mass into account, so the cylinders can be compared fairly.

(b) Change in mass = 2.40 − 2.45 = −0.05 g. Percentage change = −0.05 ÷ 2.45 × 100 = −2.04, which is −2.0 %. Check: 2.45 g decreased by 2 % is 2.45 − 0.049 = 2.401 g, which is 2.40 g.

(c) The 20 % sucrose solution was more concentrated than the cytoplasm and cell sap of the potato cells. Water moved by osmosis out of the cells, from the more dilute solution inside the cells to the more concentrated solution outside, through the partially permeable cell membranes. Losing water made the cylinder lighter.

(d) The percentage change is +6.0 % at 5 % sucrose and −2.0 % at 10 %, so zero change lies between 5 % and 10 %. The total change over this range is 8.0 percentage points, and 6.0 of these come before zero. So the estimate is 5 + (6 ÷ 8) × 5 = 8.75 %, which is about 9 %.

Topical Hub
topicalhubpages.dev